REVIEW

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# Landscape of AI literacy in education: approaches, impacts, and challenges for student preparedness—a narrative review

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**Abstract**
This narrative review explores the evolving landscape of Artificial Intelligence (AI) literacy in education. It examines current literature to identify common approaches to AI literacy development, its perceived impacts on students' technological preparedness and professional competencies, and the challenges encountered across various educational levels. The review synthesizes observations from a range of academic papers, highlighting key themes such as the importance of hands-on experience, ethical considerations, interdisciplinary strategies, and the differing needs of students from K-12 to higher and professional education. By providing an overview of existing research, this review aims to offer insights into the opportunities and complexities associated with fostering AI literacy and preparing students for an AI-driven future.

**Keywords** Artificial intelligence literacy, Technology-enhanced learning, Digital competencies, Educational innovation, Ethical AI education, Interdisciplinary pedagogy

## 1 Introduction

Artificial Intelligence (AI) has become a transformative force in society, reshaping sectors from healthcare and education to governance and impacting daily life ([6, 7]). While AI offers substantial economic benefits and novel solutions to global challenges, its rapid advancement also introduces complex ethical and societal issues. These concerns include job displacement, privacy infringement, and the potential to perpetuate systemic biases, all of which demand careful consideration of AI's impact on individual and national autonomy [9, 33, 52, 54]. The dual nature of AI necessitates the cultivation of AI literacy to ensure its responsible development and deployment.

In response, the educational sector has increasingly emphasized the development of **AI literacy**, which encompasses not only technical knowledge but also a critical understanding of AI's socio-ethical implications. Researchers have proposed frameworks that integrate affective, behavioural, cognitive, and ethical dimensions to foster this holistic competence [10, 14, 56, 57]. For example, the ED-AI Lit framework outlines six core components: Knowledge, Evaluation, Collaboration, Contextualization, Autonomy, and Ethics [5]. The overarching goal is to empower students with the critical thinking skills necessary for their future careers and to promote the responsible use of AI tools [29, 39, 44]. Although research in higher and adult education remains in its preliminary stages, a clear consensus is emerging on the need to refine AI literacy concepts and teaching methodologies [22]

Despite this growing recognition, significant gaps hinder the effective implementation of AI literacy in education. A primary challenge is the lack of a consistent definition or comprehensive framework, which complicates the establishment of coherent learning outcomes and pedagogical strategies [3, 77]. Furthermore, existing initiatives often overemphasize technical competencies while neglecting the ethical and societal considerations crucial for responsible AI use [3, 75]. This issue is compounded by a lack of teacher preparedness, as educators frequently report inadequate professional development and curricular support [61, 69]. These challenges are exacerbated by inequities in access to AI education, which threaten to deepen existing educational divides, particularly for students in under-resourced regions [51, 72, 75]. Consequently, educators struggle to meaningfully integrate emerging AI content with national standards and preexisting academic goals [58, 73].

The urgency of addressing these gaps is underscored by the complex digital landscape where students must navigate a mixture of credible information, misinformation, and disinformation, often amplified by AI [17, 26, 77]. Without a strong foundation in AI literacy, students may struggle to critically engage with AI-generated content and make informed decisions [23, 70, 79]. These interrelated challenges affirm the critical need for a more holistic and inclusive approach to AI literacy to prepare students for the demands of an AI-driven future [16, 24, 77]. Against this backdrop, this narrative literature review provides a broad overview of the current state of knowledge, aiming to identify key development approaches, assess their impact on technological preparedness, and explore how they cultivate professional competencies among students across different educational levels [58].

## 2 Theoretical framework

This research is grounded in a multidimensional theoretical framework that integrates perspectives from learning theory, developmental psychology, and educational sociology to examine how individuals acquire, apply, and internalize knowledge, skills, and values in the context of AI literacy and education, as follows:

### 2.1 Experiential and constructivist foundations

Kolb's [37] Experiential Learning Theory and Vygotsky's [74] Social Constructivism provide the foundation for understanding learning as an active, socially mediated process. Kolb emphasizes the cyclical nature of experience, reflection, conceptualization, and experimentation, while Vygotsky highlights the role of social interaction and the "zone of proximal development" in scaffolding learning. Together, these theories explain how students build knowledge through hands-on engagement and collaborative dialogue.

### 2.2 Moral and ethical development

Kohlberg's [36] Theory of Moral Development expands the framework by addressing how learners develop ethical reasoning. In the context of AI literacy, this theory underpins the importance of equipping students not only with technical skills but also with the capacity to critically evaluate the ethical and societal implications of AI technologies.

### 2.3 Situated and flexible learning

Lave and Wenger's [43] Situated Learning Theory emphasizes that learning is context-dependent and occurs within communities of practice. This aligns with Spiro et al.'s [67] Cognitive Flexibility Theory, which stresses the need for learners to adapt knowledge to complex and novel situations. Together, these perspectives support the idea that AI literacy must be developed in authentic, problem-based contexts that require transfer across domains.

### 2.4 Agency and self-regulation

Zimmerman's [80] model of Self-Regulated Learning highlights learners' active role in setting goals, monitoring progress, and employing strategies for effective learning. This perspective contributes to understanding how students can take ownership of their AI literacy development through metacognitive and motivational processes.

### 2.5 Professional socialization and career development

Weidman's [76] Career Socialization Theory situates learning within broader professional trajectories, emphasizing how educational experiences shape career identity and readiness. This is particularly relevant for AI literacy, which increasingly intersects with workforce demands across multiple disciplines.

### 2.6 Systemic and ecological perspectives

Finally, Bronfenbrenner's [13] Educational Ecosystem Theory provides a macro-level lens, situating individual learning within interconnected systems—ranging from the classroom to institutional, societal, and technological environments. This ecological perspective acknowledges that AI literacy is influenced by policy, culture, and global technological trends.

Taken together, these theories suggest that AI literacy education should be:

- Experiential and socially constructed (Kolb, Vygotsky)
- Ethically informed (Kohlberg)
- Situated and adaptable (Lave & Wenger, Spiro et al.)
- Self-directed and agentic (Zimmerman)
- Career-oriented (Weidman)
- Systemically contextualized (Bronfenbrenner)

This integrative framework underscores the importance of designing AI literacy initiatives that are learner-centred, ethically grounded, context-sensitive, and aligned with professional and societal needs.

## 3 Methodological approach

A narrative review methodology was chosen to survey the existing research landscape and identify key discussions and findings concerning AI literacy. This approach is particularly suited for synthesizing a broad, diverse, and rapidly evolving field where standardized interventions are not yet common. Unlike a systematic review, which adheres to strict, replicable search and selection protocols to facilitate meta-analysis, the goal of this narrative review is to provide a comprehensive overview, identify key themes, and discuss current trends and gaps. The synthesis presented is therefore selective and interpretive, aiming to map the conceptual territory of AI literacy in education rather than to provide a definitive quantitative summary.

To gather relevant literature, a broad search was conducted primarily using the Semantic Scholar corpus, which encompasses a vast collection of interdisciplinary academic papers. This search was facilitated by the free version of Elicit™, an AI-powered research assistant, which helped in identifying a range of literature pertinent to AI literacy. The use of these tools was justified by their capacity to perform conceptual searches across multiple disciplines, which is essential for a topic that spans computer science, education, and the social sciences. The search was guided by keywords including, but not limited to, "Artificial Intelligence literacy," "AI education," "AI competencies," "K-12 AI," and "higher education AI." The search was primarily limited to literature published between 2020 and 2024 to capture the most recent developments, particularly following the widespread availability of generative AI.

The initial search yielded approximately 400 potentially relevant academic papers. From this body of work, studies were selected for inclusion based on their direct relevance to the review's central themes: AI literacy development approaches, impacts on student preparedness, and challenges in implementation across educational levels (K-12, higher, and professional education). Papers focusing exclusively on technical AI development without an educational component or those not available in English were excluded. For the papers considered, attention was paid to extracting specific details to build a comprehensive picture of the current discourse. This included:

- Study Design Type: General classifications of research methodologies (e.g., experimental, descriptive) as reported in the papers.
- Participant Characteristics: Information about study participants, such as educational level, sample size, and discipline.
- AI Literacy Intervention Details: Descriptions of educational programs or interventions, including content, delivery methods, and AI skills taught.
- Outcome Measures: How AI literacy was assessed, including tools and metrics mentioned.
- Key Findings on AI Literacy Impact: Reported results on the perceived impact of AI literacy interventions on understanding, attitudes, and future preparedness.

The gathered information was then thematically organized by the research team to synthesize the observations and present an overview of the current state of AI literacy education as reflected in the selected literature.

## 4 Results: exploring the landscape key themes

The literature surveyed included a range of studies, examples of which are presented in Table 1, illustrating the breadth of research being conducted in AI literacy across various educational contexts and populations.

**Table 1** Studies

| Author, Year | Article Title | Study Setting | Population Type | AI Literacy Focus | Key Outcomes |
|---|---|---|---|---|---|
| [1] | Artificial Intelligence Literacy for Library Science Students: A Survey | Higher education | Master of Library Science students (n = 8) | AI knowledge areas for library science | AI literacy levels. perceptions of AI impact on library services |
| [2] | The scale of AI literacy on student learning achievement in the context of college-level education in the digital era | Higher education | University students (n = 74) | General AI literacy | Impact on academic performance |
| [4] | Assessing AI Literacy and Online Information Search Skills in Medical Education Students | Higher education | Medical education students (n = 336) | AI competencies in healthcare | AI literacy levels, online information search skills |
| [8] | Examining artificial intelligence literacy among pre-service teachers for future classrooms | Higher education | Pre-service teachers (n = 529) | AI literacy for future classrooms | AI use, detection, ethics, creation, and problem-solving |
| [11] | Cultivating Tomorrow's Innovators | Secondary education | High school students | Comprehensive AI literacy curriculum | Student engagement, comprehension |
| [12] | Day of AI: Innovating pedagogical practices to bring AI literacy to classrooms at scale | K-12 education | K-12 education | General AI literacy development | Understanding of AI concepts, ethical considerations |
| [15] | AI literacy in K-12: a systematic literature review | K-12 education | K-12 education | Comprehensive AI literacy competencies | AI knowledge, skills, and competencies across educational levels |
| [18] | Measuring Artificial Intelligence Literacy for Undergraduate Students in a Large University | Higher education | University students (n = 25) | Basic AI concepts and literacy | Improvement in AI literacy scores |
| [19] | Integrating a Module on AI Literacy in Construction Management Curriculum | Higher education | Undergraduate construction management students (n = 5) | AI literacy in construction | Understanding of AI fundamentals. application in construction |
| [19] | Enhancing AI literacy of undergraduate students using construction safety context | Higher education | Undergraduate construction students | AI literacy in construction safety | Enhancement of AI literacy levels |
| [20] | Development of an Artificial Intelligence Literacy Scale for Primary School Students | Primary education | Primary school students (n = 549) | Basic AI knowledge and skills | AI readiness. confidence, and relevance perception |
| [21] | An examination on primary students' development in AI literacy through digital story writing | Primary education | Primary students (n = 82) | AI literacy through digital story writing | Application of AI knowledge in problem-solving |
| [25] | Equipping First-Year Engineering Students with AI Literacy for Critical Evaluation of AI-Generated Content | Higher education | First-year engineering students (n = 28) | AI literacy for engineering | Understanding and critical evaluation of AI-generated content |
| [27] | Analysis of students' artificial intelligence literacy in higher education | Higher education | Computer Science students (n = 65) | General AI literacy | Understanding of AI concepts, application usage, ethical awareness |
| [28] | Examining the Impact of AI Literacy on Academic Norms and Ethics Among University Students | Higher education | University students (n = 75) | AI literacy impacts academic norms and ethics | Changes in academic norms, ethics, and AI application |

**Table 1** (continued)

| Author, Year | Article Title | Study Setting | Population Type | AI Literacy Focus | Key Outcomes |
|---|---|---|---|---|---|
| [30] | Establishing AI literacy before adopting AI | K-12 education | K-12 educators and students | General AI literacy | Increase in AI knowledge, optimism about AI benefits |
| [35] | AI literacy and teaching self-efficacy enhancement in future pre-service teachers: A case study of education office camp implementation | Secondary education | High school students | AI digital competences for future teachers | Enhancement of AI literacy and teaching efficacy |
| [34] | Examining the impact of flipped learning for developing young job seekers' AI literacy | Higher education | Adult learners (n = 80) | AI literacy for non-Computer Science majors | Improvement in AI literacy and problem-solving skills |
| [40] | AI empowerment for sustainable development: Enhancing self-efficacy and learning engagement | Secondary and higher education | Secondary school and university students (n = 224) | AI empowerment | Increase in AI empowerment and self-efficacy |
| [39] | Pedagogical delivery and feedback for an AI literacy program for university students with diverse academic backgrounds: Flipped classroom learning approach with project-based learning | Higher education | University students | AI literacy through flipped classroom | Development of AI concepts and ethical awareness |
| [41] | The development of an AI literacy framework for higher education | Higher education | University students (n = 82) | Conceptual understanding of AI | Improvement in AI literacy and conceptual understanding |
| [42] | AI literacy in global media education: Developing critical and ethical understanding | Higher education | University students (n = 29) | AI literacy in global media education | Increase in AI knowledge and awareness |
| [45, 46] | Artificial intelligence literacy in higher and adult education: A scoping literature review | Higher education | Medical students (n = 24) | AI literacy for medical imaging | Increase in AI readiness and understanding |
| [47] | Effect of a flipped classroom course to foster medical students' AI literacy with a focus on medical imaging: A single group pre-and post-test study | Higher education | Medical students (n = 377) | General AI literacy for medical students | Assessment of AI literacy levels and attitudes |
| [49] | Integrating AI literacy across disciplines: A case study in higher education | Higher education | University students (n = 300) | AI literacy across discipline | Assessment of AI knowledge, skills |
| [48] | The effect of artificial intelligence literacy education on university students' ethical consciousness of artificial intelligence | Higher education | College and university students | AI literacy and ethics | Changes in perception of AI ethics |
| [53] | Enhancing AI literacy for learning and professional practice: A design-based research approach | Higher education | Tertiary-level students | AI literacy for learning and professional practice | Development of critical skills for AI use |
| [55] | AI education and AI literacy | Primary education | Primary students (n = 82) | AI literacy through digital story writing | Application of AI knowledge in problem-solving |
| [56] | Artificial intelligence (AI) literacy education in secondary schools: A review | Secondary education | Secondary school students (n = 363) | Comprehensive AI literacy assessment | Development and validation of AI Literacy |

**Table 1** (continued)

| Author, Year | Article Title | Study Setting | Population Type | AI Literacy Focus | Key Outcomes |
|---|---|---|---|---|---|
| [57] | Design and validation of the AI literacy questionnaire: The affective, behavioral, cognitive and ethical approach | Secondary education | Secondary students | AI literacy curriculum review | Identification of AI literacy teaching approaches |
| [62] | Investigating the moderating effects of social good and confidence on teachers' intention to prepare school students for artificial intelligence education | Secondary education | Secondary school students (n = 605) | AI literacy competencies | Assessment of AI literacy across gender and school types |
| [63] | Enhancing AI literacy in higher education through integrated library services | Higher education | Undergraduate students | AI literacy integration in library services | Changes in AI literacy levels and ethical understanding |
| [64] | Artificial intelligence literacy for innovation: Enhancing skills for future workplaces | Higher education | Undergraduate and postgraduate students (n = 4) | AI literacy for innovation | Effective use of AI tools in innovation processes |
| [65] | AI-enhanced education: Exploring the impact of AI literacy on Generation Z's academic performance in Northern India | Higher education | Generation Z students | AI literacy impact on academic performance | Relationships between AI literacy, usage, and outcomes |
| [68] | AI literacy for an international technology-infused workplace | Higher education | University students | AI literacy for international technology-infused workplace | Development of AI literacy skills and cultural understanding |
| [71] | Combining human and artificial intelligence for enhanced AI literacy in higher education | Higher education | Graduate students (n = 37) | AI literacy through generative AI tools | Enhanced understanding and critical assessment of AI |
| [78] | Integrating ethics and career futures with technical learning to promote AI literacy for middle school students: An exploratory study | Secondary education | Middle school students (n = 58) | Comprehensive AI literacy (technical, ethical, career) | Improvement in AI concept understanding and ethical awareness |
| [79] | An effectiveness study of teacher-led AI literacy curriculum in K-12 classrooms | Secondary education | Middle school students (n = 25) | Teacher-led AI literacy curriculum | Improvement in AI concept understanding and career awareness |

### 4.1 AI literacy development approaches

A synthesis of the reviewed literature reveals a strong pedagogical consensus favouring active and contextualized learning approaches. As detailed in Table 2, methodologies such as hands-on experience, project-based learning, and flipped learning are consistently highlighted. This trend suggests a departure from passive instruction towards models that require learners to directly engage with AI tools and concepts. Notably, the integration of ethical considerations is not treated as a standalone topic but is frequently woven into these applied contexts, indicating a growing recognition that technical skill and ethical reasoning are inseparable components of AI literacy.

The reviewed literature highlights several recurring approaches to fostering AI literacy. A strong emphasis is placed on practical, hands-on experiences, where learners directly engage with AI tools and applications to build applied understanding. Equally important is the integration of ethical considerations, ensuring that students not only develop

**Table 2** AI Literacy Development Approaches

| Theme | Observations from the Literature | Illustrative Studies | Implementation Context |
|---|---|---|---|
| Practical, hands-on experience | Hands-on experience with AI tools appears to enhance understanding and engagement | Cheng et al. [19], Feldman and Cherry [25], Sengewald and Tremmel [64] | Undergraduate and postgraduate levels, various disciplines |
| Integration of ethical considerations | Ethical awareness is frequently highlighted as a crucial component of AI literacy | Kong et al. [38], Lee [48], Okolo [59] | Across educational levels, emphasis in social science disciplines |
| Interdisciplinary approach | AI literacy seems to benefit from integration across various disciplines | Irfan et al. [32], Cheng et al. [19], Sorte et al. [66] | Journalism, construction, medicine |
| Project-based learning | Project-based approaches are suggested to enhance AI concept understanding and application | Kong et al. [38], Zhang et al. [78] | University level, middle school |
| Flipped learning | The flipped classroom approach has been reported to improve AI literacy for diverse student groups | Kim et al. [34] | Adult learners, both Computer Science (CS) and non-CS majors |

technical proficiency but also cultivate critical awareness of the societal and moral implications of AI. An interdisciplinary orientation emerges as another key theme, reflecting the need to connect AI literacy across multiple fields rather than confining it to computer science alone. In addition, project-based learning and flipped learning approaches are frequently reported, both of which encourage active student participation, collaboration, and deeper engagement with AI concepts.

The contexts in which these approaches were implemented varied across educational levels and disciplines. Several studies examined AI literacy at both undergraduate and postgraduate levels, spanning a wide range of subjects. Others focused more narrowly, exploring applications within Social Sciences, or targeting specific professional domains such as journalism, construction, and medicine**.** Beyond traditional university settings, AI literacy initiatives were also found in middle school programs**,** adult learning contexts, and in comparative studies contrasting the experiences of Computer Science (CS) majors and non-CS majors**.**

Collectively, these observations indicate that AI literacy development is multifaceted, blending technical, ethical, and interdisciplinary dimensions, while adapting to diverse educational and professional contexts.

### 4.2 Technological preparedness

The literature suggests that AI literacy interventions contribute to the development of durable, transferable skills that extend beyond proficiency with specific tools. As summarized in Table 3, recurring outcomes include enhanced problem-solving and improved critical thinking, particularly in evaluating AI-generated content. This pattern implies that the primary value of AI literacy may lie in fostering higher-order cognitive skills, equipping students with the adaptability needed to navigate future, as-yet-unknown technological shifts.

Across the reviewed studies, four interrelated themes consistently emerged in relation to technological preparedness**:** enhanced problem-solving skills, improved critical thinking, increased adaptability to emerging technologies, and practical proficiency with AI tools**.**

**Table 3** Impact on Technological Preparedness

| Theme | Observations from the Literature | Illustrative Studies | Implementation Context |
|---|---|---|---|
| Enhanced problem-solving skills | AI literacy is often linked to improvements in problem-solving abilities in technological contexts | Kong et al. [39], Ayanwale et al. [8], Kim et al. [34] | Secondary school, pre-service teachers, adult learners |
| Improved critical thinking | AI literacy is suggested to foster critical evaluation of AI-generated content and technologies | Feldman and Cherry [25], Olanipekun [60], Zhang et al. [79] | Undergraduate engineering, media education, middle school |
| Increased technological adaptability | AI literacy is reported to enhance students' ability to adapt to new technologies | Singh et al. [65], Hollands and Breazeal [30], Chen and Zhang [18] | Generation Z students, K-12 educators, higher education |
| AI tool proficiency | Students are described as developing practical skills in using AI tools relevant to their field | Sengewald and Tremmel [64], Tzirides et al. [45, 46, 71]) | Undergraduate and postgraduate, graduate students, medical students |

The strongest evidence comes from higher education studies, where multiple sources report that AI literacy initiatives can significantly enhance students' problem-solving abilities when integrated into technological contexts. This finding appears well-supported, though some studies note variation depending on discipline and instructional design.

Evidence for the role of AI literacy in fostering critical thinking is somewhat more mixed. While several studies highlight positive outcomes, especially in prompting students to evaluate AI-generated content critically, other sources suggest that this effect may be contingent on explicit pedagogical emphasis rather than an automatic outcome of exposure to AI tools.

The theme of technological adaptability is widely discussed but supported by fewer empirical studies. Those that do exist emphasize students' ability to transfer knowledge and skills across contexts, suggesting a growing but still emergent line of inquiry. Similarly, while reports of practical AI tool proficiency are present across fields such as medicine, journalism, and education, the strength of evidence varies, with some findings drawn from small-scale case studies rather than large comparative analyses.

The contexts in which these themes were investigated highlight notable diversity. While a substantial proportion of studies focus on undergraduate learners, others extend into secondary schools, K–12, middle school, pre-service teacher education, graduate programs, and professional domains such as media and medicine. This breadth suggests a growing recognition of AI literacy as a cross-cutting competence, though the uneven distribution of evidence across educational stages indicates areas in need of further study.

Taken together, the literature reveals both consensus and gaps: there is broad agreement that AI literacy contributes to problem-solving and critical engagement with technology, yet questions remain regarding how consistently these outcomes can be achieved across contexts. The variability in methodological rigor and the limited number of large-scale, longitudinal studies underscore the need for cautious interpretation and for future research to strengthen the empirical base.

### 4.3 Professional competency enhancement

Beyond general technological skills, AI literacy is framed in the literature as a critical component of professional readiness for an AI-integrated workforce. The themes in Table 4 show a dual focus: preparing students for the future of work in general through

**Table 4** Professional competency enhancement

| Theme | Observations from the Literature | Illustrative Studies | Implementation Context |
|---|---|---|---|
| Career readiness | AI literacy is linked to improved awareness of AI's impact on future careers | Zhang et al. [78], Zhang et al. [79], Kim and Nam [35] | Middle school, high school |
| Discipline-specific AI competencies | AI literacy is reported to enhance professional skills in specific disciplines | Irfan et al. [32], Cheng et al. [19], Sorte et al. [66] | Journalism, construction, medicine |
| Ethical decision-making in professional contexts | AI literacy is suggested to improve the ability to navigate ethical challenges in AI use | Lee [48], Olanipekun [60], [45, 46]) | Undergraduate, media education, medical students |
| Collaborative AI skills | AI literacy is described as enhancing the ability to work collaboratively with AI tools | Sanusi et al. [62], Tzirides et al. [71] | Secondary school, graduate students |

early career awareness and developing highly specific AI competencies tailored to fields like medicine, construction, and journalism. This highlights a key trend: the shift from a monolithic concept of AI literacy to a more nuanced understanding of domain-specific AI literacies.

The reviewed studies highlight four primary themes concerning AI literacy in professional settings**:** career readiness, discipline-specific AI competencies, ethical decision-making in professional contexts, and collaborative AI skills**.**

Career readiness is supported by several studies conducted in middle and high school environments. These consistently suggest that early exposure to AI literacy increases students' awareness of AI's implications for their future career trajectories. While the evidence base is relatively robust in secondary education, fewer longitudinal studies trace whether this preparedness translates into actual career outcomes.

Discipline-specific competencies appear frequently in fields such as journalism, construction, and medicine. Here, the evidence is moderately strong, with multiple studies documenting how AI literacy training strengthens domain-specific skills. However, the narrow disciplinary focus raises questions about transferability to other fields.

Ethical decision-making emerges as a widely emphasized theme, particularly across undergraduate education, media studies, and medical training. While many studies underscore the importance of equipping students with tools to navigate algorithmic bias and responsible AI use, the strength of empirical evidence varies. Some works are conceptual or exploratory, whereas others provide case-based demonstrations, indicating a growing but uneven evidence base.

Collaborative AI skills are less frequently studied but remain noteworthy. A small number of studies—primarily at the secondary and graduate education levels—report that collaborative tasks involving AI tools foster teamwork and adaptability. The limited quantity of research, however, tempers the generalizability of these claims.

Taken together, the literature points to a developing consensus: AI literacy can enhance professional preparedness by cultivating career awareness, ethical judgment, technical competence, and collaboration. Yet, the evidence base is uneven, with stronger support for career readiness and ethical awareness, and comparatively weaker, more exploratory evidence for collaboration. Key gaps include the lack of longitudinal tracking and limited comparative studies across professional domains.

**Table 5** Long-term Impact Indicators

| Impact Area | Potential Success Factors (Examples from individual studies) | Potential Challenges (Examples from individual studies) | Suggested Best Practices (Examples from individual studies) |
|---|---|---|---|
| Technological Adaptability | Hands-on experience with AI tools, Interdisciplinary approach | Rapid evolution of AI technology, Varying levels of prior knowledge | Regular curriculum updates, Personalized learning paths |
| Ethical Awareness | Integration of ethical considerations in technical learning, Real-world case studies | Complexity of ethical issues, Balancing technical and ethical content | Collaborative ethical problem-solving, Ongoing ethical discussions |
| Career Preparedness | Exposure to AI career possibilities, Industry partnerships | Rapidly changing job market, Limited real-world AI experience | Internships and industry projects, Career counselling with AI focus |

### 4.4 Cross-educational level differences and long-term impact

The approaches and intended impacts of AI literacy are not uniform but are strategically differentiated across educational levels, from foundational concepts in K-12 to specialized applications in professional education. This differentiation suggests an emerging, albeit informal, developmental trajectory for AI literacy.

K-12 Education: An emphasis is often placed on foundational AI concepts, ethical awareness, and career exploration. Research by Zhang et al. [78] and Zhang et al. [79] suggested that middle school students can comprehend basic AI principles and reflect on their future implications.

Undergraduate Education: Literature points to the integration of advanced technical concepts with practical applications. Studies such as Cheng et al. [19] and Irfan et al. [32] illustrated how discipline-specific AI literacy can foster the development of professional competencies.

Graduate and Professional Education: An emphasis on specialized applications and ethical considerations is often noted. Findings from Tzirides et al. [71] and [45, 46]) highlighted the role of advanced AI literacy in equipping students to navigate professional challenges.

Pre-Service Teacher Education: Multiple studies underscored the critical need to prepare future educators for integrating AI literacy into their teaching practices. Hur [31] and [45, 46] identified unique pedagogical requirements of this group in AI literacy training as reported in their work.

However, as the synthesis of long-term indicators in Table 5 reveals, the field still faces significant challenges in translating these educational interventions into lasting and measurable impacts. The literature identifies numerous potential success factors and challenges, yet there is little consensus or replication of findings, indicating that the long-term effectiveness of AI literacy education remains a critical area for future investigation.

Overall, the evidence suggests that AI literacy education is shaped by a heterogeneous mix of enabling factors and persistent challenges**.** There is a degree of consensus around recurring challenges, particularly the tension between technical and ethical demands, but less agreement on universally effective success factors. The current evidence base is exploratory and fragmented, underscoring the need for more systematic, comparative, and longitudinal studies to identify which strategies are most impactful across contexts and how they interact with key dimensions of technological preparedness, ethical reasoning, and professional competencies.

## 5 Discussion

The observations from the reviewed literature underscore the perceived importance of practical engagement, ethical considerations, interdisciplinary learning, and adaptive teaching strategies in AI literacy education. In terms of emerging pedagogical approaches, practical, hands-on experience was frequently discussed as a valuable method. This suggests that experiential learning may foster deeper engagement with AI concepts and technologies. Project-based learning and flipped learning were also mentioned as supporting active learning strategies, emphasizing student-centred instruction. The integration of ethical considerations across multiple studies highlights a growing recognition in the literature that AI literacy extends beyond technical proficiency to include responsible and informed usage. The interdisciplinary nature of AI literacy education, as noted in several papers, suggests its relevance across a broad spectrum of disciplines, reinforcing the idea that customized approaches based on field-specific requirements may be beneficial.

Regarding technological preparedness, the impact of AI literacy appears significant across various educational levels according to the authors reviewed. Enhanced problem-solving skills and critical thinking are recurring themes, indicating that AI literacy may foster analytical abilities that extend beyond AI-specific applications. Increased technological adaptability and AI tool proficiency further reinforce the potential role of AI literacy in preparing students for rapidly evolving technological landscapes. Notably, the variation in implementation contexts described, ranging from secondary school to postgraduate education, suggests that AI literacy is considered beneficial across different cognitive and professional development stages.

AI literacy education is also portrayed in the literature as playing a crucial role in professional preparedness, as evidenced by discussions around career readiness, discipline-specific competencies, ethical decision-making, and collaborative AI skills. The emphasis on career readiness at middle and high school levels in some studies underscores a perceived need to introduce AI literacy early to inform career aspirations and future employability. Additionally, the identification of discipline-specific AI competencies in certain papers highlights a potential necessity for tailored curricula that align with industry requirements. Ethical decision-making, examined across multiple professional domains, points to the importance attributed to training individuals to navigate the ethical dilemmas associated with AI. Collaborative AI skills suggest that AI literacy may not only enhance individual competencies but also foster teamwork in AI-integrated environments.

The literature also emphasizes distinct AI literacy needs at different educational levels. For instance, K-12 AI education is often described as focusing on foundational AI concepts, ethical considerations, and career exploration. Undergraduate AI education appears to integrate technical skills with practical applications, while graduate and professional AI education emphasizes specialization and ethical decision-making. Pre-service teacher AI education emerges as a crucial area in some discussions, highlighting the need to equip future educators with AI literacy competencies to prepare their students for an AI-driven world. These variations underscore the idea that adapting AI literacy curricula to align with learners' cognitive development, professional aspirations, and ethical considerations at each stage of education is likely necessary.

A critical examination of the literature reveals a potential disconnect between the macro-level societal problems that AI literacy purports to address, such as workforce disruption and algorithmic inequity, and the micro-level outcomes typically measured in educational interventions. While studies report success in improving students' conceptual understanding or tool proficiency, there is a notable gap in evidence demonstrating that these gains translate into the resilient, critical, and ethical competencies needed to navigate the complex societal transformations wrought by AI. Future research must bridge this gap by developing longitudinal studies and more authentic assessment methods that can measure the real-world impact of AI literacy education on civic engagement, professional ethics, and lifelong adaptability. Furthermore, the literature reveals tensions and conflicting findings, such as AI's potential to support multilingual learning while simultaneously reinforcing dominant linguistic norms like standardized English.

The recurring theme of teacher unpreparedness highlights a central paradox in AI professional development. While many initiatives focus on addressing first-order barriers by providing resources and training on current AI tools, the rapid evolution of the technology renders such training ephemeral. The literature suggests that a more sustainable approach may lie in addressing second-order barriers, namely, teachers' pedagogical beliefs and attitudes. Fostering a pedagogy of critical inquiry and adaptability, rather than tool-specific proficiency, could better equip educators to integrate not just current AI, but any future technological advancement, into their practice. This points to a need for a fundamental shift in professional development from functional training to fostering deep pedagogical resilience.

The patterns observed in the literature resonate with various theoretical perspectives. For instance, the consistent emphasis on hands-on experience and project-based learning aligns with Kolb's [37] Experiential Learning Theory. These observations suggest that AI literacy education may be most effective when learners actively engage with AI tools in applied contexts. This reinforces the notion that AI education could prioritize interactive and experiential approaches over passive instruction.

Also, the interdisciplinary nature of AI literacy education suggests that constructivist learning theories, such as Vygotsky's [74] Social Constructivism, could be crucial for effective learning. Some studies highlight the importance of collaborative and contextualized learning experiences, where students construct knowledge through interactions with peers, educators, and AI systems in various disciplines.

On the other hand, the integration of ethical considerations in AI literacy education, as seen in the literature, resonates with Kohlberg's (1981) Theory of Moral Development. Some findings indicate that AI literacy programs might play a critical role in fostering students' ability to critically evaluate ethical dilemmas associated with AI technologies, suggesting that moral reasoning could be a core component of AI education frameworks.

Furthermore, the findings related to discipline-specific AI competencies can be viewed through the lens of Lave and Wenger's [43] Situated Learning Theory, which posits that knowledge acquisition is most effective when embedded within the relevant professional or academic context. The diverse applications of AI literacy in journalism, medicine, and construction noted in some papers suggest that AI education might be best tailored to the unique demands of each field.

Additionally, the identification of AI literacy as a potential driver of technological adaptability aligns with Spiro et al.'s (1988) Cognitive Flexibility Theory. The ability of students and professionals to navigate and adapt to AI-driven technological landscapes, as described in some literature, suggests that AI education could emphasize flexible thinking and transferable knowledge across multiple domains.

Also, the single study on flipped learning mentioned supports the potential role of self-regulated learning (Zimmerman, 1989) in AI literacy education. By shifting the responsibility of knowledge acquisition to the learners, flipped learning models might enhance students' engagement and deepen their understanding of AI concepts. This suggests that future AI literacy curricula could explore incorporating more self-directed and autonomous learning strategies.

Furthermore, the observations related to career readiness suggest that AI literacy education aligns with Career Socialization Theory. Some results indicate that exposure to AI tools and ethical decision-making in professional contexts helps students develop essential career competencies, supporting the argument that AI literacy could be integrated into workforce development programs.

And last, but no less important, the success factors and challenges identified in AI literacy education suggest a complex interplay of cognitive, technological, and ethical dimensions that align with the Ecological Systems Theory [13], which posits that learning outcomes are influenced by multiple interconnected factors, including institutional support, industry partnerships, and policy frameworks.

## 6 Conclusion

Based on the synthesized literature, emerging findings suggest that AI literacy programs may benefit from prioritizing active learning methodologies. The patterns observed in the literature point toward the potential value of incorporating experiential learning opportunities, such as hands-on AI tool applications and real-world problem-solving projects, to enhance students' comprehension and retention. The reported effectiveness of flipped learning and project-based learning in AI education in some studies suggests that institutions might explore the adoption of innovative teaching methods to enhance student engagement. A combination of self-directed learning and applied in-class activities could facilitate deeper understanding and skill acquisition.

Given the consistent emphasis on ethics in the reviewed studies, it appears crucial for educational programs to systematically integrate discussions on algorithmic bias, responsible AI decision-making, and the broader social impact of AI technologies. This approach could ensure that students develop a critical understanding of the ethical challenges associated with AI. Likewise, the literature reviewed highlights a potential need for AI literacy education beyond computer science disciplines. AI literacy could be incorporated into fields such as social sciences, medicine, journalism, and construction to facilitate cross-disciplinary applications and enhance its relevance across diverse professional sectors.

Moreover, the correlation noted in some studies between AI literacy and career preparedness underscores the potential necessity of developing workforce skills, thus aligning educational programs with labour market demands. This could entail embedding discipline-specific AI competencies within curricula, fostering adaptability to emerging AI technologies, and equipping students with proficiency in AI tools relevant to their respective industries. For its part, the variations in AI literacy approaches across different educational stages, as observed in the literature, highlight a potential need for tailored instructional strategies. For example:

- In K-12 Education: Emphasize foundational AI concepts, ethical considerations, and career exploration.
- In Undergraduate Education: Integrate advanced technical knowledge with real-world applications to develop professional competencies.
- In Graduate and Professional Education: Focus on specialized AI applications and ethical decision-making in professional contexts.
- In Pre-Service Teacher Education: Prepare future educators to incorporate AI literacy into their teaching methodologies, addressing both pedagogical integration and ethical concerns.

In summary, the literature suggests that AI literacy programs should be adaptable, interdisciplinary, and aligned with workforce requirements, ensuring that learners develop both technical competencies and a nuanced understanding of AI's societal implications. The observations gathered in this review contribute to the growing discourse on AI literacy by identifying key pedagogical strategies, perceived impacts on technological preparedness, and professional applications as discussed in the current academic landscape. Future research could continue to explore how AI literacy programs can be continuously adapted to keep pace with evolving AI technologies and workforce demands. Also, further studies on assessment methodologies could enhance the understanding of how to effectively measure AI literacy outcomes across diverse learning contexts.

## 7 Limitations

This review has several limitations that should be considered when interpreting its findings. First, as a narrative review, it provides a broad overview of the AI literacy landscape but does not employ the exhaustive, systematic search and screening protocols of a systematic review. Consequently, the synthesis is selective by nature, and it is possible that relevant studies were not included.

Second, the search was primarily conducted using the Semantic Scholar database via Elicit™. While this provides access to a vast collection of academic literature, it does not cover all possible databases, and the exclusion of other sources may have limited the scope of the included work.

Finally, the identification of themes is an interpretive process conducted by the research team. While grounded in the literature, this thematic analysis is inherently subjective and represents one possible synthesis of the available evidence. These limitations underscore that the findings should be viewed as a snapshot of key trends and discussions in a rapidly evolving field, rather than a definitive or exhaustive account.

**Author contributions**

GV: Conceptualization, project administration, methodology, formal analysis, writing—original draft, and supervision.CR: Literature Review, Resources, Writing—Review & Editing.LG: Literature Review, Resources, Writing—Review & Editing.

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**Data availability**

The data supporting the results and analysis of this paper is publicly available and can be found in The Semantic Scholar Open Research Corpus (S2ORC), a general-purpose corpus for NLP and text mining research over scientific papers built and maintained by Semantic Scholar's research team, which indexes over two hundred million academic papers. Papers are aggregated into a unified source to create the largest publicly available collection of machine-readable academic text, provided as a JSON archive.

## Declarations

**Ethics approval and consent to participate**
Not applicable.

**Consent for publication**
Not applicable.

**Competing interests**
The authors declare no competing interests

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**References**

1. Abid M, Ben-Salha O, Kanetaki Z, Sekrafi H. Does the impact of artificial intelligence on unemployment among people with disabilities differ by educational level? IEEE Access. 2024. https://doi.org/10.1109/ACCESS.2024.3456962.
2. Agusnaya N, Nirmala P. The scale of AI literacy on student learning achievement in the context of college-level education in the digital era. J Vocat Inform Computer Educ. 2024. https://doi.org/10.61220/voice.v2i2.20243.
3. Akgün S, Greenhow C, Willet KBS. Artificial intelligence in education: addressing ethical challenges in K–12 settings. AI & Soc. 2021;36(1):1–13. https://doi.org/10.1007/s00146-020-00993-6.
4. Ali MA, Mughari S. Library and information sciences students' digital literacy as a predictor of their academic performance. J Media Horizons 2025;6(2):1054–66.
5. Allen LK, Kendeou P. ED-AI lit: an interdisciplinary framework for AI literacy in education. Policy Insights Behav Brain Sci. 2024;11(1):3–10. https://doi.org/10.1177/23727322231220339.
6. Alon-Barkat S, Busuioc M. Human–AI interactions in public sector decision making:"automation bias" and "selective adherence" to algorithmic advice. J Public Adm Res Theory. 2023;33(1):153–69.
7. Alowais SA, Alghamdi SS, Alsuhebany N, Alqahtani T, Alshaya AI, Almohareb SN, et al. Revolutionizing healthcare: the role of artificial intelligence in clinical practice. BMC Med Educ. 2023;23(1):689.
8. Ayanwale MA, Adelana OP, Molefi RR, Adeeko O, Ishola A. Examining artificial intelligence literacy among pre-service teachers for future classrooms. Computers Educ Open. 2024. https://doi.org/10.1016/j.caeo.2024.100179.
9. Babashahi L, Barbosa CE, Lima Y, Lyra A, Salazar H, Argôlo M, et al. AI in the workplace: a systematic review of skill transformation in the industry. Adm Sci. 2024;14(6):127.
10. Bewersdorff, A., Nerdel, C., & Zhai, X. 2023. *How AI literacy correlates with affective, behavioral, cognitive and contextual variables: A systematic review*. Technical University of Munich. https://www.edu.sot.tum.de/fileadmin/w00bed/edu/DigiLLab/Documents/Bewersdorff_Nerdel_Zhai_Review_AI_Literacy_Correlations_preprint.pdf
11. Bosarge E. Cultivating tomorrow's innovators: Navigating the landscape of high school AI literacy. In: 2024 ASEE Annual Conference & Exposition; 2024.
12. Breazeal, C., Du, X., Abelson, H., Klopfer, E., & Park, H. W. (n.d.). Day of AI: Innovating pedagogical practices to bring AI literacy to classrooms at scale.n. https://doi.org/10.1007/978-981-99-7947-9_20
13. Bronfenbrenner U. The ecology of human development: Experiments by nature and design. Harvard University Press; 1979.
14. Carolus A, Koch M, Straka S, Wienrich C. MAILS — Meta AI literacy scale: development and testing of an AI literacy questionnaire based on well-founded competency models and psychological change- and meta-competencies. Computers Human Behav Artif Humans. 2023. https://doi.org/10.1016/j.chbah.2023.100014.
15. Casal-Otero L, Catala A, Fernández-Morante C, Taboada M, Cebreiro B, Barro S. AI literacy in K-12: a systematic literature review. Int J STEM Educ. 2023;10(1):29. https://doi.org/10.1186/s40594-023-00418-7.
16. Chaudhry, A., Cukurova, M., & Luckin, R. A transparency index framework for AI in education. *arXiv Preprint*. 2022. https://arxiv.org/abs/2206.03220
17. Chu-Ke C. Misinformation and literacies in the era of generative artificial intelligence: abrief overview and a call for future research. J Commun Inq. 2024. https://doi.org/10.1177/27523543241240285.
18. Chen YH, Zhang K. Impact of basic artificial intelligence (AI) course on understanding concepts, literacy, and empowerment in the field of AI among students. Comput Appl Eng Educ. 2025;33(1):e22806. https://doi.org/10.1002/cae.22806.
19. Cheng J-Y, Devkota A, Gheisari M, Jeelani I. Enhancing AI literacy of undergraduate students using construction safety context. Constr Res Congress. 2024. https://doi.org/10.1061/9780784485293.005.
20. Dai Y, Chai CS, Lin PY, Jong MSY, Guo Y, Qin J. Promoting students' well-being by developing their readiness for the artificial intelligence age. Sustainability. 2020;12(16):6597. https://doi.org/10.3390/su12166597.
21. Davy NGTK, Luo W, Chan H, Chu S. An examination on primary students' development in AI literacy through digital story writing. Computers Educ Artif Intell. 2022. https://doi.org/10.1016/j.caeai.2022.100054.
22. Dewan U, Hingle A, McDonald N, Johri A. Engineering educators' perspectives on the impact of generative AI in higher education. In: 2025 IEEE Global Engineering Education Conference (EDUCON). IEEE; 2025. pp. 1–10. https://doi.org/10.48550/arXiv.2502.00569.
23. EdWeek. "We're at a disadvantage," and other teacher sentiments on AI. Education Week. 2024. https://www.edweek.org/technology/were-at-a-disadvantage-and-other-teacher-sentiments-on-ai/2024/10.
24. FairAIED. Navigating fairness, bias, and ethics in educational AI applications. *arXiv Preprint*. 2024. https://arxiv.org/abs/2407.18745

25. Feldman U, Cherry C. Equipping first-year engineering students with artificial intelligence literacy (AI-L): Implementation, assessment, and impact. In: 2024 ASEE Annual Conference & Exposition; 2024.
26. Fulsher A, Pagkratidou M, Kendeou P. GenAI and misinformation in education: A systematic scoping review of opportunities and challenges. AI & Society. 2025;1–13. https://doi.org/10.1007/s00146-025-01706-4.
27. Hamsar I, Dewantara H, Andini H, Ardiansyah S, Afrizah NA, Getri M, et al. Analisis literasi artificial intelligence mahasiswa pada perguruan tinggi. J Vocat Inform Computer Educ. 2024. https://doi.org/10.61220/voice.v2i1.31.
28. Haris H, Muhammad RD, Arsyanda M, Wahyudi MR, Ilham M. Analisis dampak literasi artificial intelligence terhadap perubahan norma dan etika akademik mahasiswa. J Pendidik Terap. 2024;2(1):66–77. https://doi.org/10.61255/jupiter.v2i1.200.
29. Harrison L. Technicians as teachers for student experience. Pedagogy J: Learn Teach Enhancement Centre (Salford). 2025;1(1). https://doi.org/10.57898/PEDAGOGY.280.
30. Hollands FM, Breazeal C. Establishing AI literacy before adopting AI. Sci Teach. 2024. https://doi.org/10.1080/00368555.2024.2308316.
31. Hur JW. Fostering AI literacy: overcoming concerns and nurturing confidence among pre-service teachers. Inf Learn Sci. 2024. https://doi.org/10.1108/ILS-11-2023-0170.
32. Irfan M, Murray L, Ali S. Integration of artificial intelligence in academia: a case study of critical teaching and learning in higher education. Glob Soc Sci Rev. 2023. https://doi.org/10.31703/gssr.2023(VIII-I).32.
33. Khogali HO, Al-Habaibeh A, Abdo H, Alquraan T. The blended future of automation and AI: examining some weighty features of long-term AI societal impacts. Futures. 2023;145:103065. https://doi.org/10.1016/j.futures.2023.103065.
34. Kim H-J, So H, Suh Y. Examining the impact of flipped learning for developing young job seekers' AI literacy. Int Conf Artif Intell Educ. 2023. https://doi.org/10.1007/978-3-031-36272-9_77.
35. Kim M-Y, Nam J-M. AI literacy and teaching self-efficacy enhancement in future pre-service teachers: a case study of education office camp implementation. Soc Cogn Enhance Interv. 2024. https://doi.org/10.21197/JCEI.15.3.10.
36. Kohlberg, L. The psychology of moral development: The nature and validity of moral stages. Harper & Row; 1984.
37. Kolb DA. Experiential learning: experience as the source of learning and development. Prentice Hall; 1984.
38. Kong SC, Cheung WMY, Zhang G. Evaluating artificial intelligence literacy courses for fostering conceptual learning, literacy and empowerment in university students: refocusing to conceptual building. Comput Hum Behav Rep. 2022;7:100223. https://doi.org/10.1016/j.chbr.2022.100223.
39. Kong SC, Cheung MYW, Tsang O. Developing an artificial intelligence literacy framework: evaluation of a literacy course for senior secondary students using a project-based learning approach. Comput Educ: Artif Intell. 2024;6:100214. https://doi.org/10.1016/j.caeai.2024.100214.
40. Kong SC, Yang Y, Hou C. Examining teachers' behavioural intention of using generative artificial intelligence tools for teaching and learning based on the extended technology acceptance model. Comput Educ: Artif Intell. 2024;7:100328. https://doi.org/10.1016/j.caeai.2024.100328.
41. Kong, S., He, H., Zhang, G., & Cheung, M. Pedagogical delivery and feedback for an artificial intelligence literacy programme for university students with diverse academic backgrounds: Flipped classroom learning approach with project-based learning. 2022.
42. Korte SM, Cheung WMY, Maasilta M, Kong SC, Keskitalo P, Wang L, Gu MM. Enhancing artificial intelligence literacy through cross-cultural online workshops. Comput Educ Open. 2024;6:100164. https://doi.org/10.1016/j.caeo.2024.100164.
43. Lave J, Wenger E. Situated learning: Legitimate peripheral participation. Cambridge University Press; 1991.
44. Laflamme AS, Bruneault F. Redefining academic integrity in the age of generative artificial intelligence: the essential contribution of artificial intelligence ethics. J Sch Publ. 2025;56(2):481–509. https://doi.org/10.3138/jsp-2025-001.
45. Laupichler MC, Aster A, Schirch J, Raupach T. Artificial intelligence literacy in higher and adult education: a scoping literature review. Computers Educ Artif Intell. 2022;3:100101. https://doi.org/10.1016/j.caeai.2022.100101.
46. Laupichler MC, Hadizadeh D, Wintergerst M, von der Emde L, Paech D, Dick EA, et al. Effect of a flipped classroom course to foster medical students' AI literacy with a focus on medical imaging: a single group pre-and post-test study. BMC Med Educ. 2022. https://doi.org/10.1108/JHOM-04-2024-0154.
47. Laupichler MC, Aster A, Meyerheim M, Raupach T, Mergen M. Medical students' AI literacy and attitudes towards AI: a cross-sectional two-center study using pre-validated assessment instruments. BMC Med Educ. 2024;24(1):401. https://doi.org/10.1186/s12909-024-04749-1.
48. Lee A. The effect of artificial intelligence literacy education on university students' ethical consciousness of artificial intelligence. Robotics & AI Ethics 2021;6(3):52–61.
49. Lee SS, Moore RL. Harnessing generative AI (GenAI) for automated feedback in higher education: a systematic review. Online Learn. 2024;28(3):82–106. https://doi.org/10.24059/olj.v28i3.4325.
50. Li, C., Lu, G., & He, X. Measuring artificial intelligence literacy of pre-service teachers at a university in Northwest China. *Proceedings of the 2023 Twelfth International Conference of Educational Innovation Through Technology (EITT)*. 2023. https://doi.org/10.1109/EITT61659.2023.00027
51. Madaio, M., Stark, L., Wortman Vaughan, J., & Wallach, H. Beyond "fairness": Structural (in)justice lenses on AI for education. *arXiv Preprint*. 2021. https://arxiv.org/abs/2105.08847
52. Maiti M, Kar AK, Pani AK, Dutta A. A study on ethical implications of artificial intelligence. Future Bus J. 2025;11(1):12. https://doi.org/10.1186/s43093-025-00462-5.
53. Mohammadi Y, Malik S, Shibani A. Students as collaborative partners: embedding AI literacy in higher education. ASCILITE Publ. 2024;125–126. https://doi.org/10.14742/ascilite2024.1658.
54. Moses, O. *The dark side of artificial intelligence: Ethical dilemmas, job displacement, and the risk of bias.* ResearchGate. 2025. https://www.researchgate.net/publication/392159314
55. Ng DTK, Leung JKL, Su MJ, Yim IHY, Qiao MS, Chu SKW. AI education and AI literacy. In: AI literacy in K-16 classrooms. Cham: Springer International Publishing; 2022. p. 9–19. https://doi.org/10.1007/978-3-031-18880-0.
56. Ng D, Su J, Leung J, Chu SK. Artificial intelligence (AI) literacy education in secondary schools: a review. Interact Learn Environ. 2023. https://doi.org/10.1080/10494820.2023.2255228.
57. Ng D, Wu W, Leung J, Chiu T, Chu SK. Design and validation of the AI literacy questionnaire: the affective, behavioural, cognitive and ethical approach. Br J Educ Technol. 2023. https://doi.org/10.1111/bjet.13411.

58. Ng DTK, Wu W, Leung JKL, Chiu TKF, Chu SKW. Opportunities, challenges and school strategies for integrating generative AI in education. Computers Educ Artif Intell. 2025;7:100268. https://doi.org/10.1016/j.caeai.2025.100268.
59. Okolo CT. Beyond AI hype: a hands-on workshop series for enhancing AI literacy in middle and high school students. Res Equitable and Sustain Part Eng Comput Technol. 2024. https://doi.org/10.1145/3653666.3656075.
60. Olanipekun SO. Computational propaganda and misinformation: AI technologies as tools of media manipulation. World J Adv Res Rev. 2025;25(1):911–23. https://doi.org/10.30574/wjarr.2025.25.1.12345.
61. Riggs M. Impact of AI literacy professional development on teaching. STEM Educ. 2025;6(2):33–45.
62. Sanusi IT, Ayanwale MA, Chiu TK. Investigating the moderating effects of social good and confidence on teachers' intention to prepare school students for artificial intelligence education. Educ Inf Technol. 2024;29(1):273–95. https://doi.org/10.1007/s10639-023-12250-1.
63. Sayed A, Telbany S, Ashmawy S. AI-driven transformation: Advancing information literacy at the British University in Egypt library. Cybrarians J. 2024;73:62–87
64. Sengewald T, & Tremmel. Teaching AI literacy for innovation: Work in progress. In: Proceedings of the 2024 Computers and People Research Conference; 2024. pp. 1–4.
65. Singh E, Vasishta P, Singla A. Ai-enhanced education: exploring the impact of AI literacy on Generation Z's academic performance in Northern India. Qual Assur Educ. 2024. https://doi.org/10.1108/QAE-02-2024-0037.
66. Sorte SR, Rawekar A, Rathod SB. Understanding AI in healthcare: perspectives of future healthcare professionals. Cureus. 2024;16(8):e66285. https://doi.org/10.7759/cureus.66285.
67. Spiro RJ, Feltovich PJ, Jacobson MJ, Coulson RL. Cognitive flexibility, constructivism, and hypertext: random access instruction for advanced knowledge acquisition in ill-structured domains. In: Duffy TM, Jonassen DH, editors. Constructivism and the technology of instruction: A conversation. Lawrence Erlbaum; 1992. pp. 57–75.
68. Swartz S. Preparing graduates for the 21st century workplace: a transdisciplinary approach to information technology-related study programs. Interdiscip J Partnersh. 2024;11(2). https://doi.org/10.24926/ijps.v11i2.5652.
69. Tan H, Zhang Y, Liu M. Artificial intelligence in teaching and teacher professional development: challenges and opportunities. Computers Educ Artif Intell. 2024;5:100245. https://doi.org/10.1016/j.caeai.2024.100245.
70. Tsang SJ. Integrating AI literacy. J Media Lit Educ. 2025;17(2):53–65. https://digitalcommons.uri.edu/jmle/vol17/iss2/4/.
71. Tzirides A, Zapata GC, Kastania NP, Saini AK, Castro V, Ismael SAR, et al. Combining human and artificial intelligence for enhanced AI literacy in higher education. Computers Educ Open. 2024. https://doi.org/10.1016/j.caeo.2024.100184.
72. UNESCO. Artificial intelligence in education. United Nations Educational, Scientific and Cultural Organization. 2023. https://www.unesco.org/en/digital-education/artificial-intelligence
73. U.S. Department of Education. *Artificial intelligence and the future of teaching and learning: Insights and recommendations*. 2023. https://www.ed.gov/sites/ed/files/documents/ai-report/ai-report.pdf
74. Vygotsky LS. Mind in society: The development of higher psychological processes. Harvard University Press; 1978.
75. Walter Y. Embracing the future of artificial intelligence in the classroom: the relevance of AI literacy, prompt engineering, and critical thinking in modern education. Int J Educ Technol High Educ. 2024;21(1):15. https://doi.org/10.1186/s41239-024-00448-3.
76. Weidman JC. Undergraduate socialization: a conceptual approach. In: Smart JC, editor. Higher Education: Handbook of Theory and Research, Vol. 5. Agathon Press; 1989. pp. 289–322.
77. Wiese C, Hansen J, Graesser A. Ethics education in AI: a systematic review. Computers Educ Artif Intell. 2025;7:100272. https://doi.org/10.1016/j.caeai.2025.100272.
78. Zhang H, Lee IA, Ali S, DiPaola D, Cheng Y, Breazeal C. Integrating ethics and career futures with technical learning to promote AI literacy for middle school students: an exploratory study. Int J Artif Intell Educ. 2022. https://doi.org/10.1007/s40593-022-00293-3.
79. Zhang H, Lee I, Moore KS. An effectiveness study of teacher-led AI literacy curriculum in K-12 classrooms. AAAI Conf Artif Intell. 2024. https://doi.org/10.1609/aaai.v38i21.30380.
80. Zimmerman BJ. Becoming a self-regulated learner: an overview. Theory Pract. 2002;41(2):64–70. https://doi.org/10.1207/s15430421tip4102_2.


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